

Post-Quantum Cryptography

for the Novice, the Enjoyer, the Deployer and the Academic

Lena Heimberger

BalCCon 2026, Novi Sad

Cryptography is already a part of your life



messages

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messages



payments

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your car

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your ID

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every connection

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every connection

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It's a skillset, not a separate function.

Mosca's inequality: How long until my secret is revealed?

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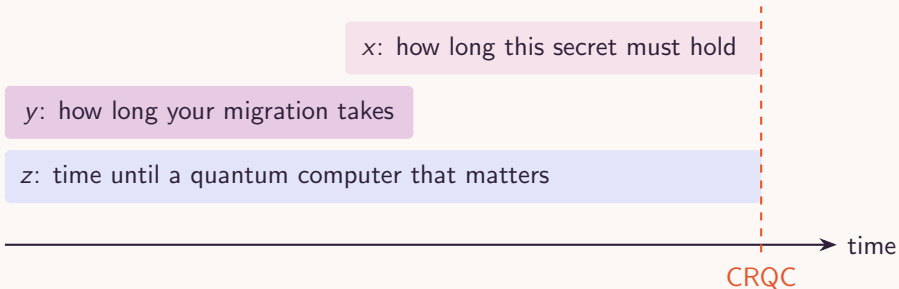
x : how long this secret must hold

Mosca's inequality: How long until my secret is revealed?

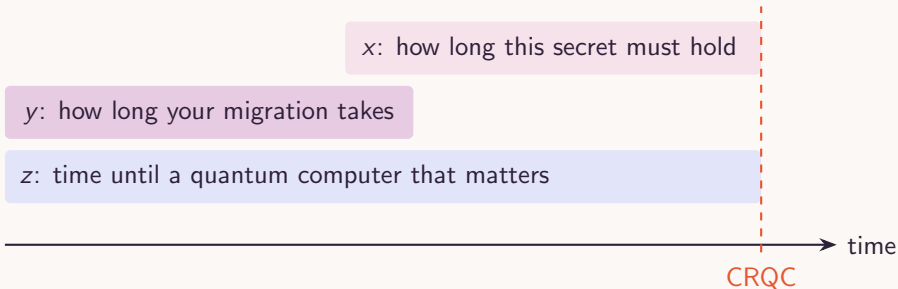
x : how long this secret must hold

y : how long your migration takes

Mosca's inequality: How long until my secret is revealed?



Mosca's inequality: How long until my secret is revealed?



If $x + y > z$, you are already too late.

M. Mosca. *Cybersecurity in an Era with Quantum Computers: Will We Be Ready?* IEEE Security & Privacy 16(5), 2018.

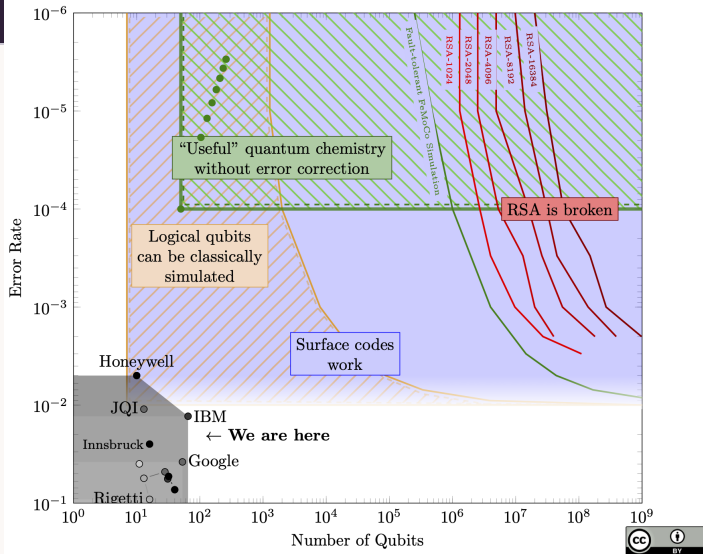


Figure 1: https://sam-jaques.appspot.com/quantum_landscape

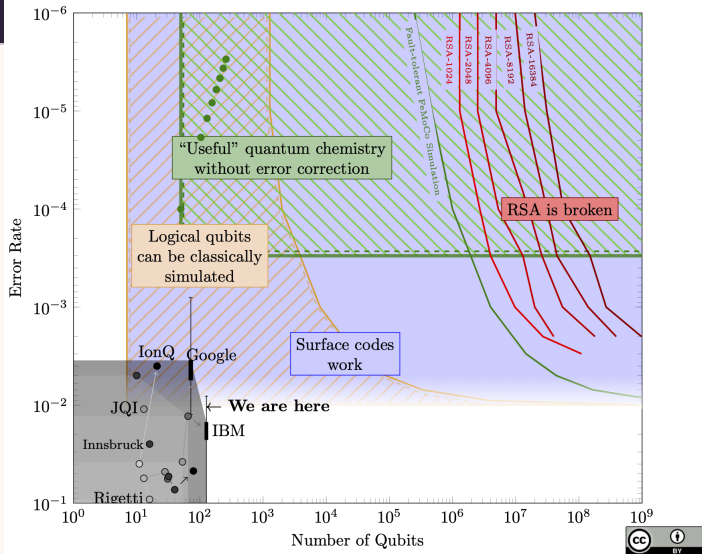


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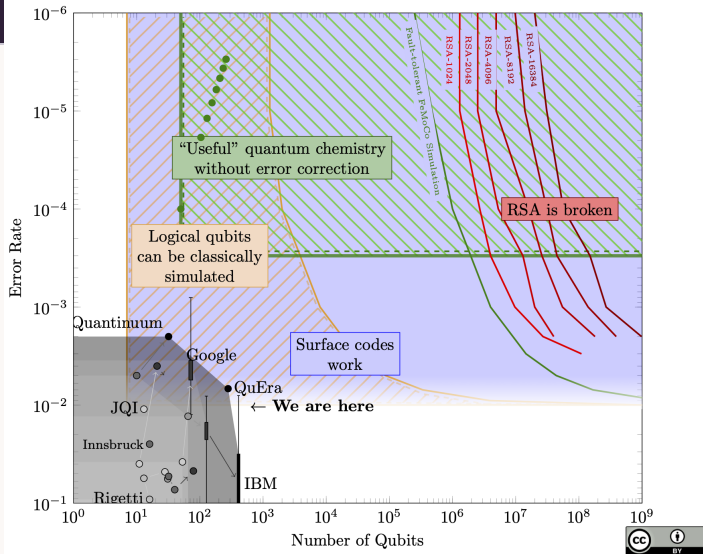
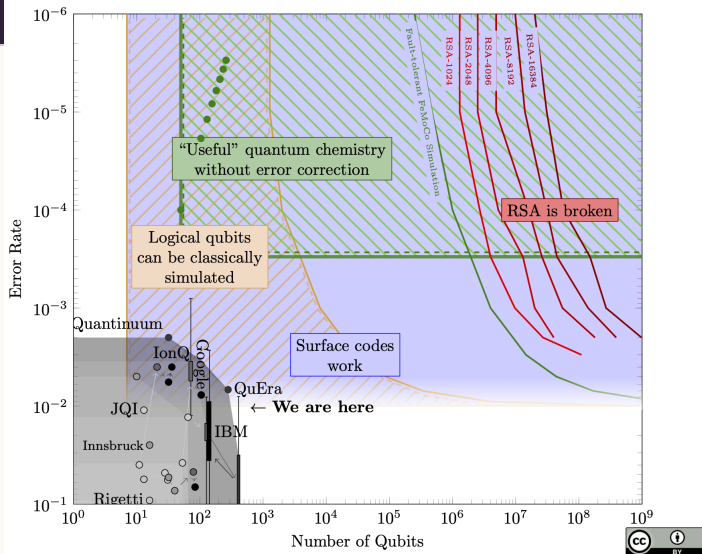


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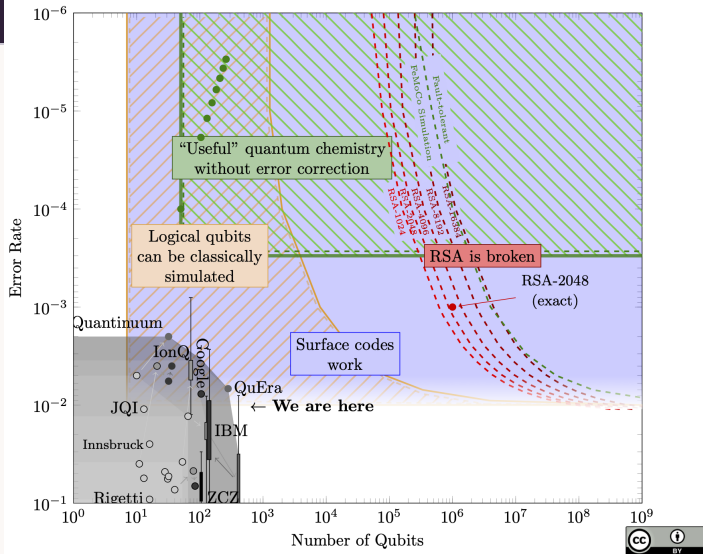
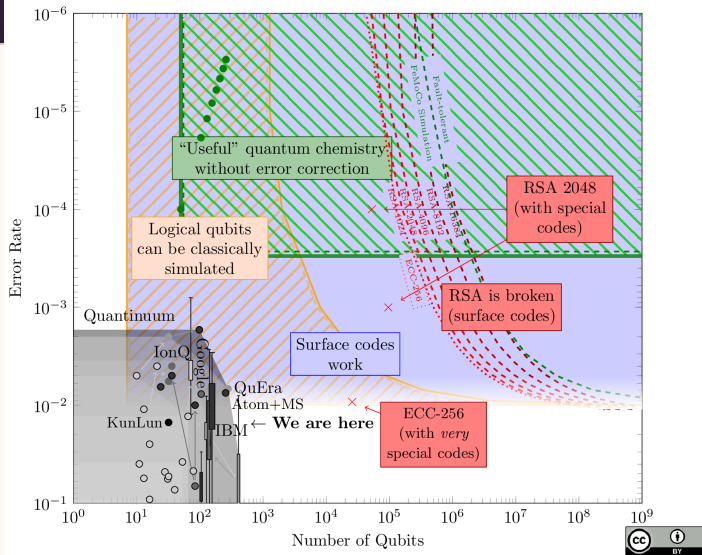
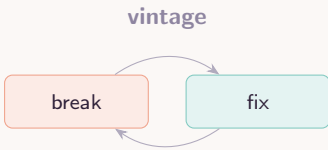


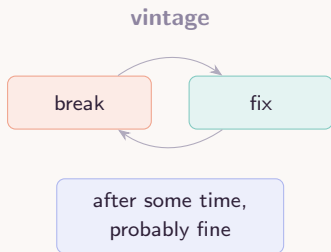
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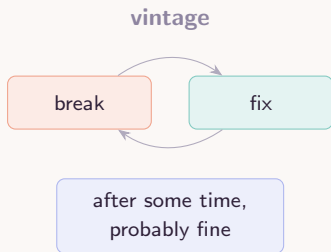
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show breaking the scheme is as hard as breaking underlying assumption

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show breaking the scheme is as hard as breaking underlying assumption
- Reason about model, tightness of reduction, which problem.

Security Levels

NIST level	at least as hard as
1	recovering an AES-128 key
2	finding a SHA-256 collision
3	recovering an AES-192 key
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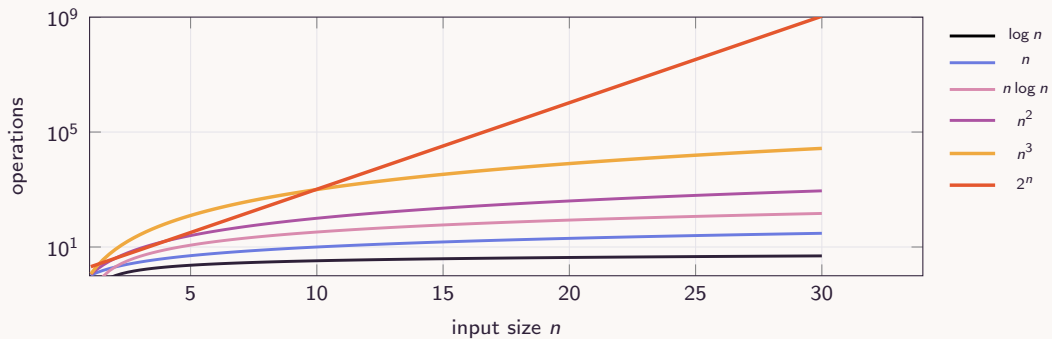
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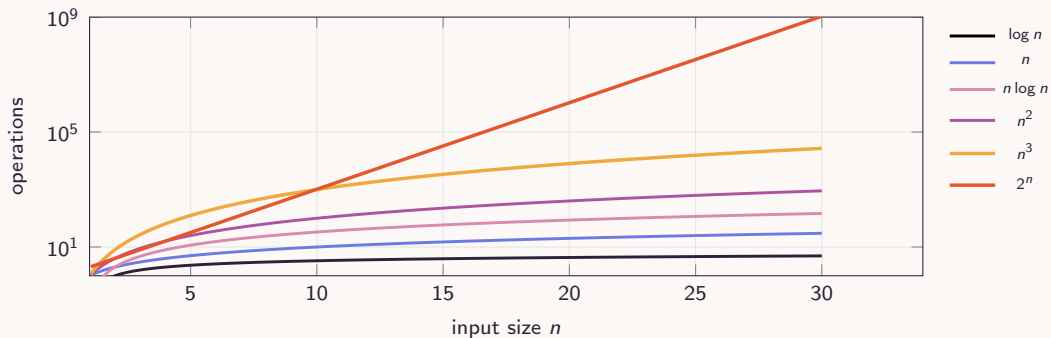
How to factor 2048 bit RSA integers with less than a million noisy qubits, Craig Gidney,

<https://arxiv.org/pdf/2505.15917>

Just double the parameters

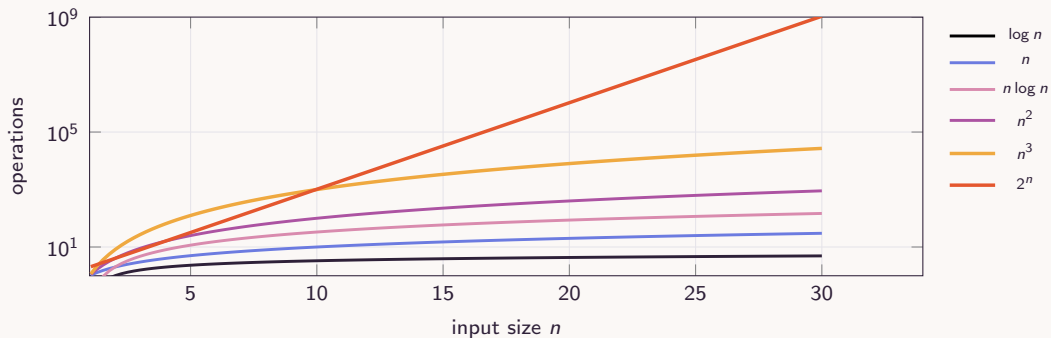


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Also doubles the work you need to do!

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Terabytes of RSA keys

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I also did applied crypto last two summers



[Privacy](#) [Terms](#)

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Talk touches a lot of points. If you ever zone out, just zone back in later.

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Signatures

Forgery with a CRQC.

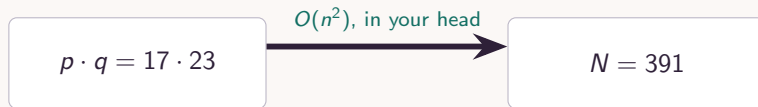
deadline: the day you stop accepting it



the Novice

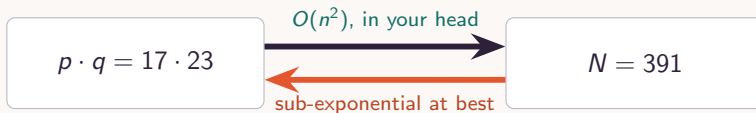
“Why does a quantum computer break cryptography?”

A hard problem becomes easy with a friend secret



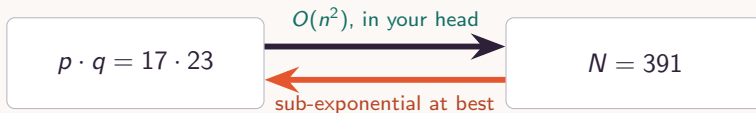
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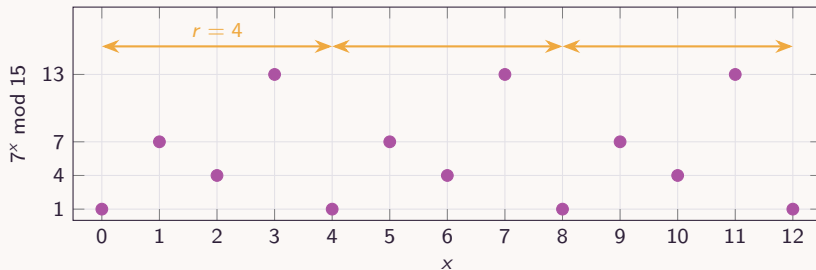
- Polynomial complexity for multiplication.
- general number field sieve runs in $\exp(O((\log N)^{1/3}(\log \log N)^{2/3}))$.
- NP-hard: easy to verify, hard to search.

That bounds the *worst* instance, and cryptography needs the *average* instance to be hard.

Shor reduces factoring to frequency finding

factor N

P. Shor, 1994/1997.

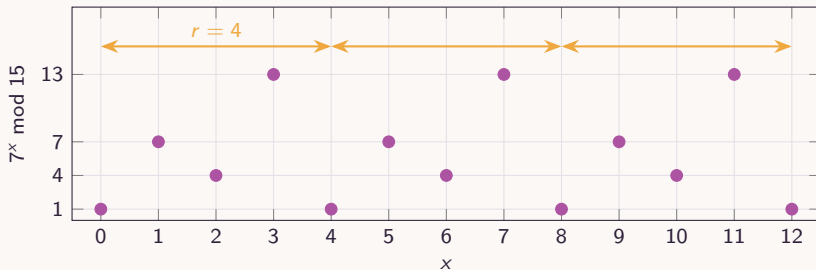


RSA security depends on the attacker not knowing the order of the group.

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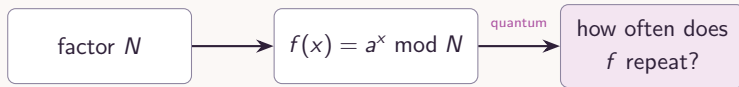


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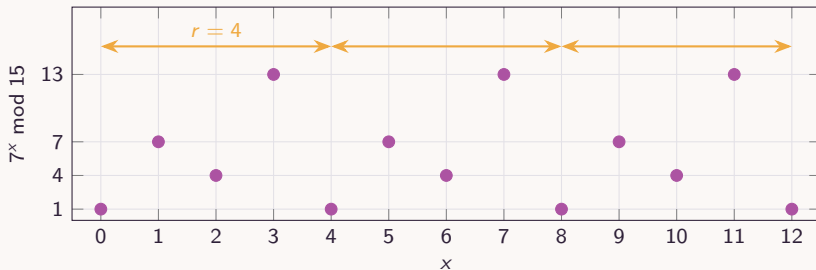


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Blinding operations

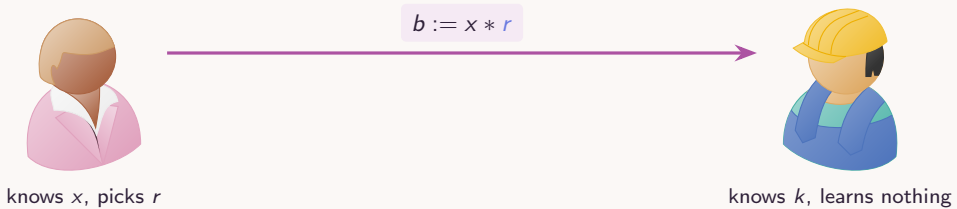


knows x , picks r

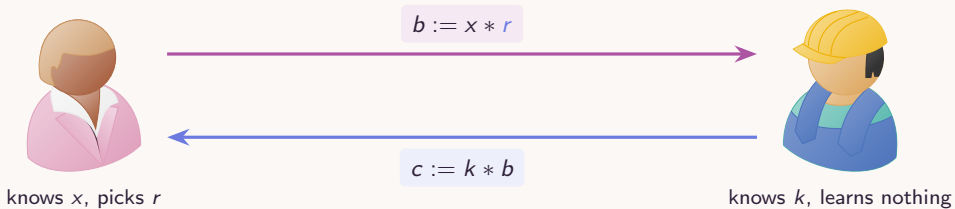


knows k , learns nothing

Blinding operations



Blinding operations



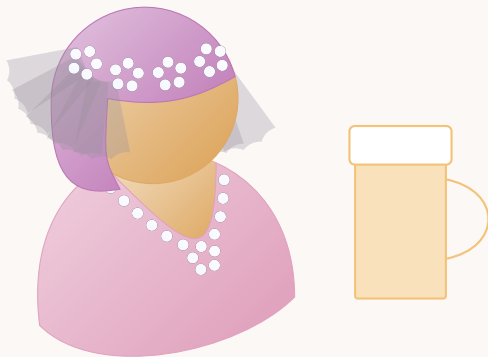
Hidden shifts in commutative settings: Kuperberg



- Same commutative structure, secret moved from *period* to *offset*.
- Then quantum computers have **subexponential** advantage.

G. Kuperberg. A subexponential-time quantum algorithm for the dihedral hidden subgroup problem.
SIAM J. Comput. 35(1), 2005.

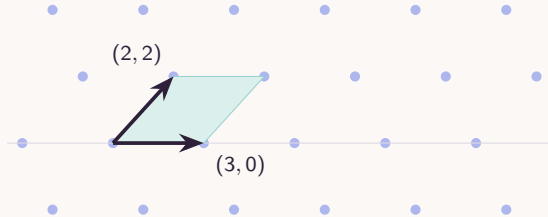
Remove the frequency.



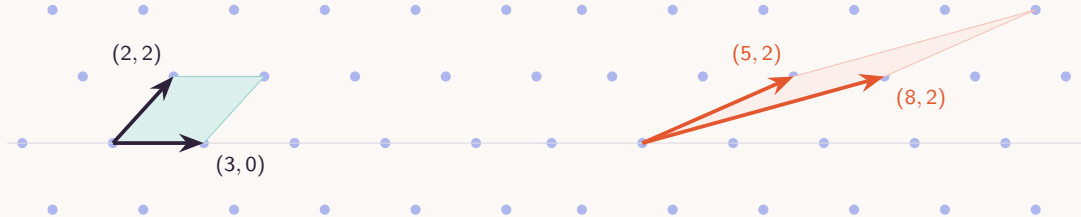
the Enjoyer

“What Alice did last night.”

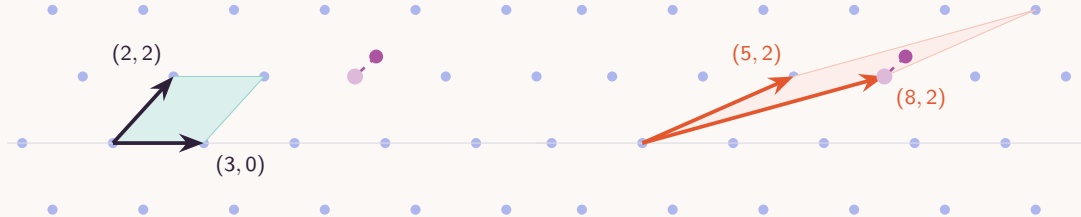
The same lattice, described twice



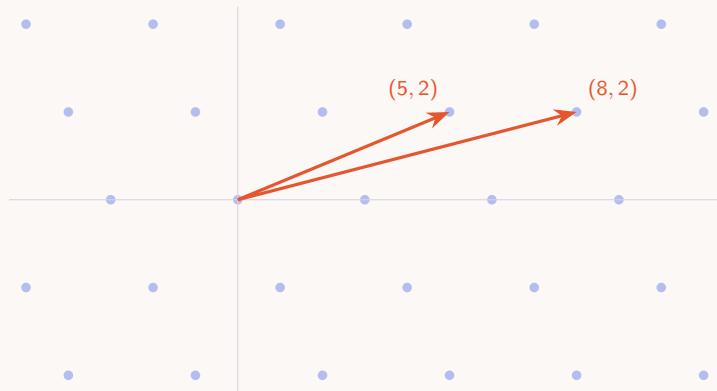
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All about that base



repeat

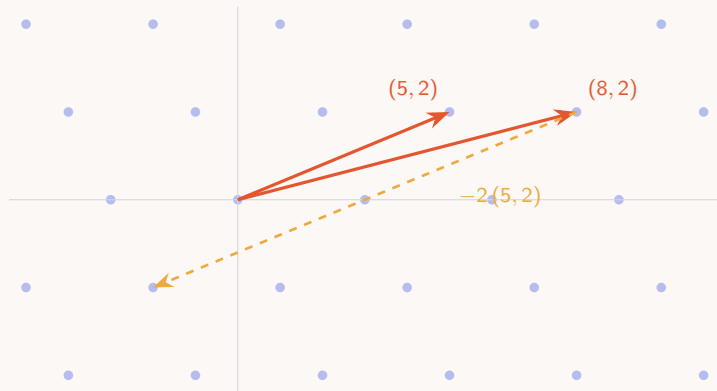
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$$b_2 \leftarrow b_2 - \mu b_1$$

swap if $\|b_2\| < \|b_1\|$

until no swap

All about that base



repeat

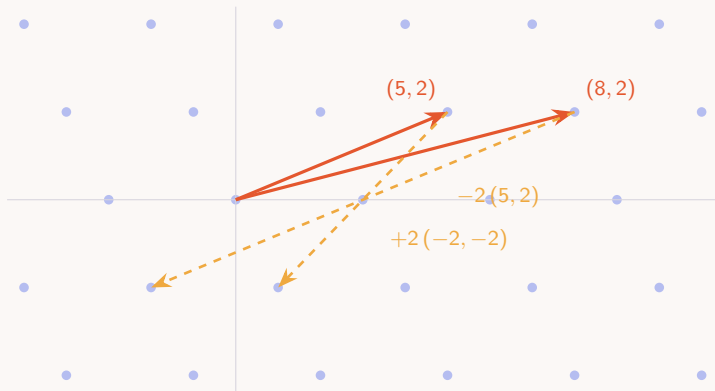
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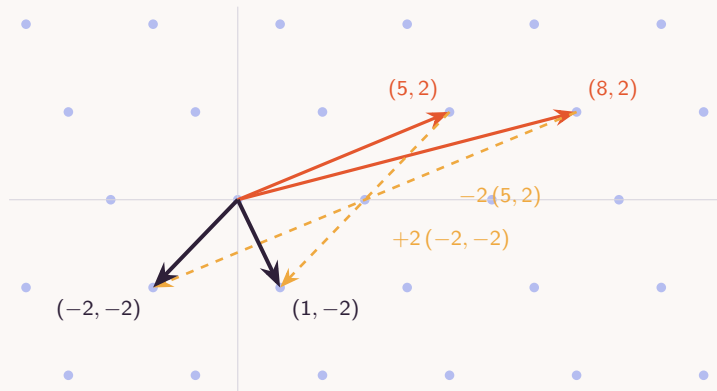
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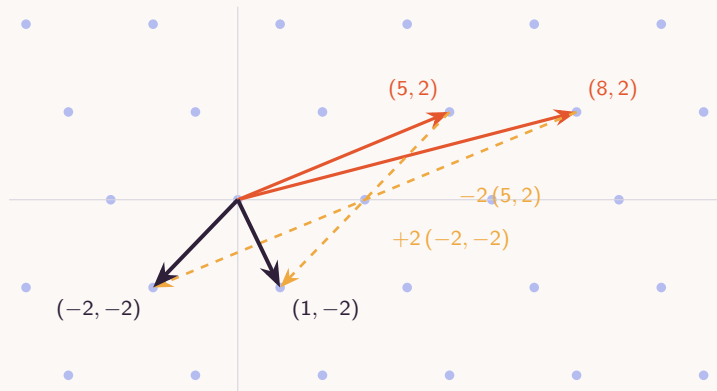
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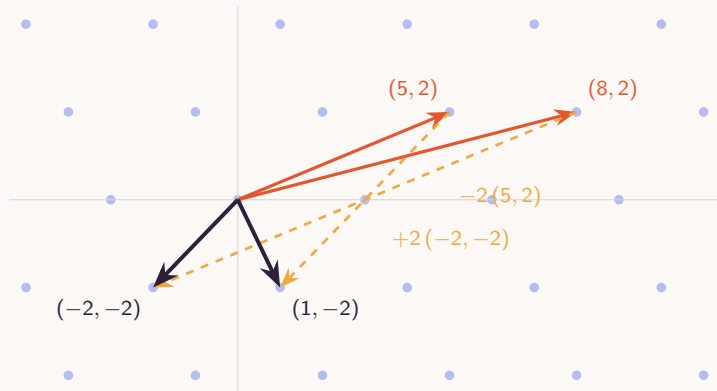
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- LLL is the approximate version.

Lattices: linear algebra with some noise

A diagram illustrating the equation $As = b + e$. The matrix A is represented by a large light blue square. The vector s is a tall, narrow light purple rectangle. The vector e is a tall, narrow light orange rectangle. The vector b is a tall, narrow light teal rectangle. The equation is shown as A followed by s , then a plus sign, then e , then an equals sign, then b . Below each box is descriptive text: A is "public, uniform, big coefficients"; s is "secret, short"; e is "noise, short"; and b is "published".

$$A s + e = b$$

public, uniform,
big coefficients

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Lattices: linear algebra with some noise

A diagram illustrating the equation $A \cdot s + e = b$. The terms are represented by colored boxes: a large light blue square for A , a tall narrow purple rectangle for s , a tall narrow orange rectangle for e , and a tall narrow teal rectangle for b . The equation is written as $A \cdot s + e = b$ with the boxes placed around the operators. Below each box is descriptive text: "public, uniform, big coefficients" for A , "secret, short" for s , "noise, short" for e , and "published" for b .

$$A \cdot s + e = b$$

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- Given A and b , find s .

Lattices: linear algebra with some noise

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Drop e and this is Gaussian elimination.

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rings and modules to shrink the key, rounding for determinism.

A well-studied problem

- system of univariate linear equations over integers with some noise

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Rise from the ashes of cryptanalysis.

Codes: the same equations, counted differently

sent	1	0	1	1	0	0	1	0	1	1	0	1
received	1	0	0	1	0	0	1	1	1	1	0	1

two bits flipped

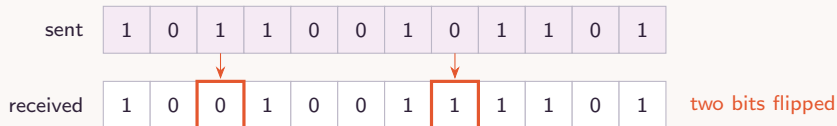
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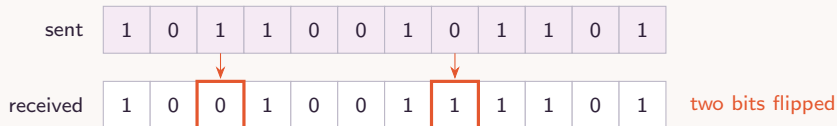
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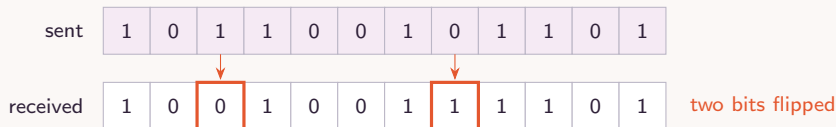
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Sparse solution instead of a small one.

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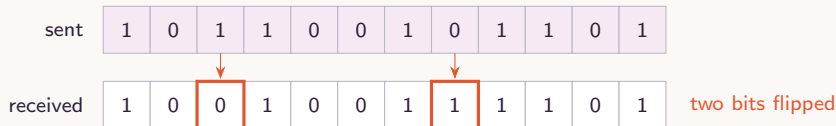
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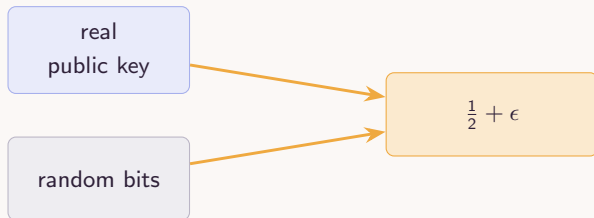
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- ~~conservative choice~~. Distinguisher attack.

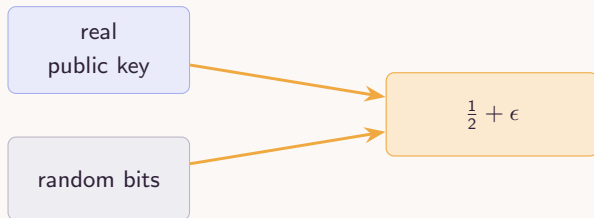
What a distinguisher actually means



Quasipolynomial Cryptanalysis of the McEliece Cryptosystem (or: PIR Meets McEliece)

Ghoshal, Ishai, Jain, Sun, ia.cr/2026/1630

What a distinguisher actually means

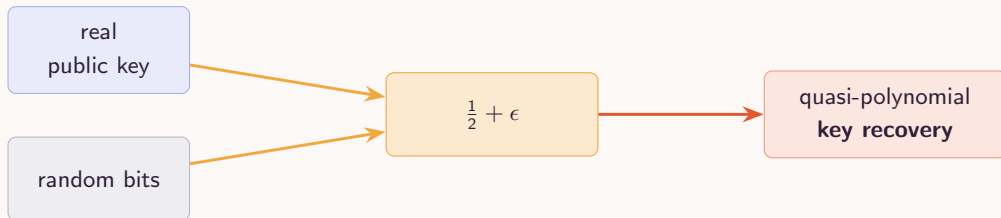


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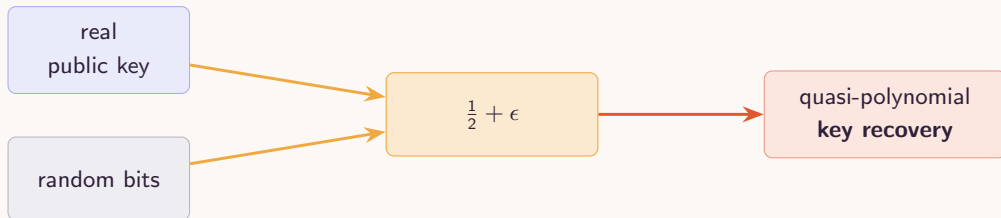


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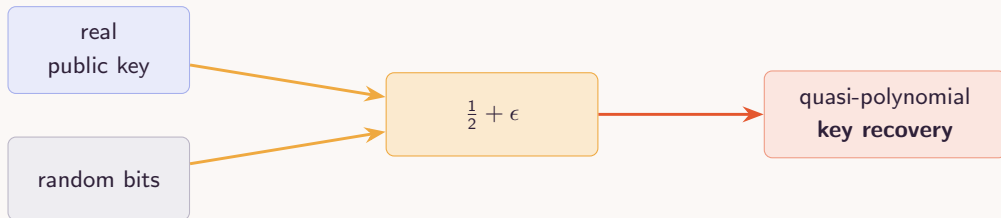


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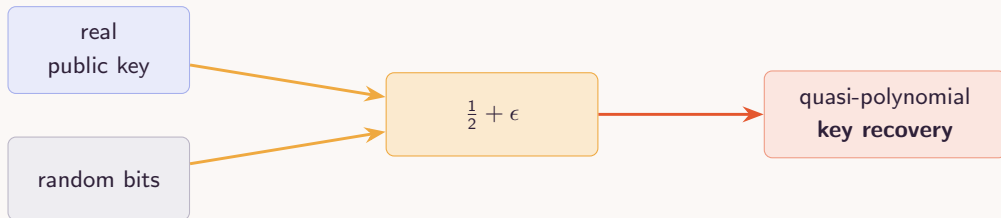


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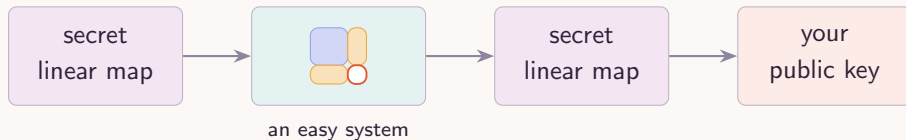


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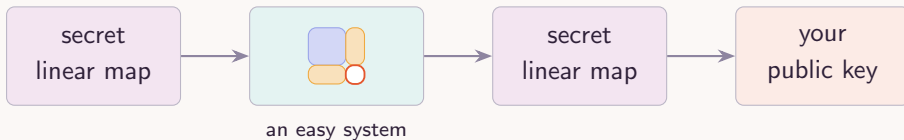
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Multivariate: hide an easy system inside a hard-looking one

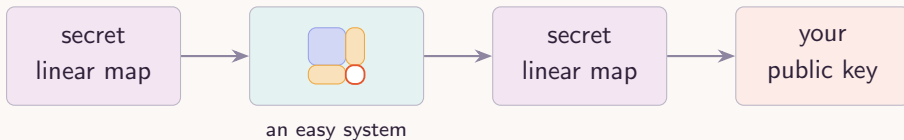


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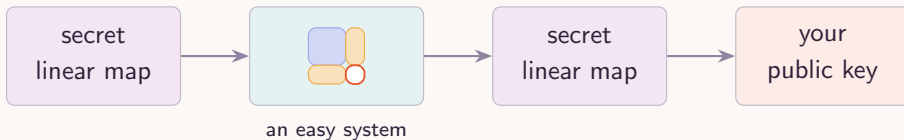
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- 96-byte signatures, public keys in the tens of kilobytes

The easy system: oil and vinegar

vinegar	oil
$v \times v$	$v \times o$
$o \times v$	0

no oil times oil: fix
the vinegar and the
rest is *linear*

■ sig: 96 bytes

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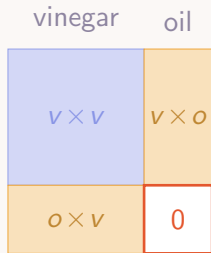
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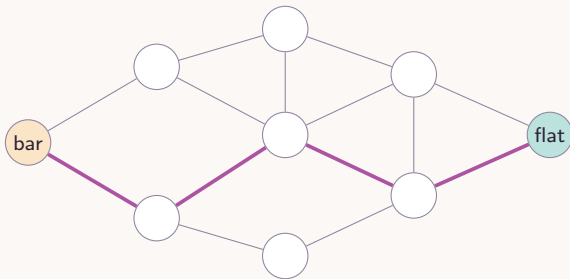
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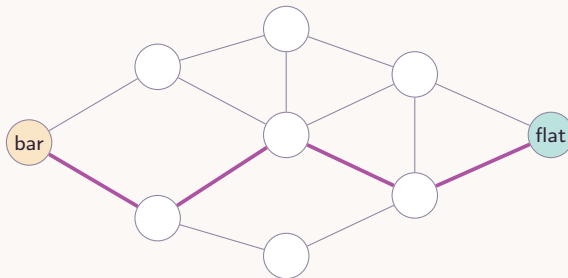
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- **Balanced** oil and vinegar died in 1998 (Kipnis-Shamir). **Unbalanced**, 2 – 3× more vinegar, still standing
- Shrinking the public key with more structure is risky

Alice walks home alone at night



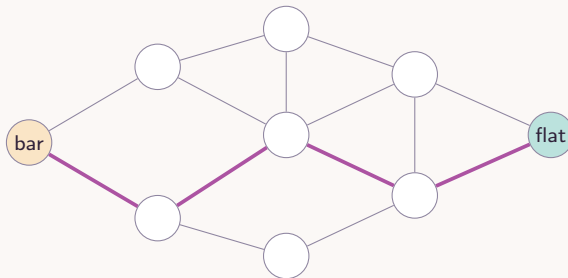
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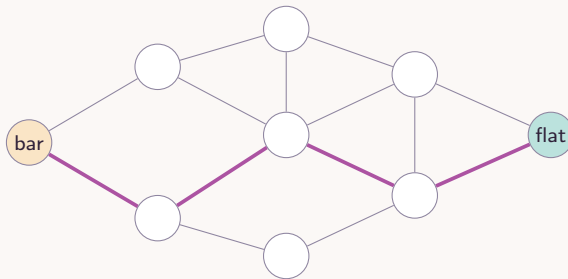
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- Start: Bar, End: Got home safe text
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- Nodes are elliptic curves, steps are maps between them: Isogeny walk!

Isogenies: small, elegant, and a lot of work

- SQIsign: 148-byte signatures

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Public key and signature together fit in a tweet.

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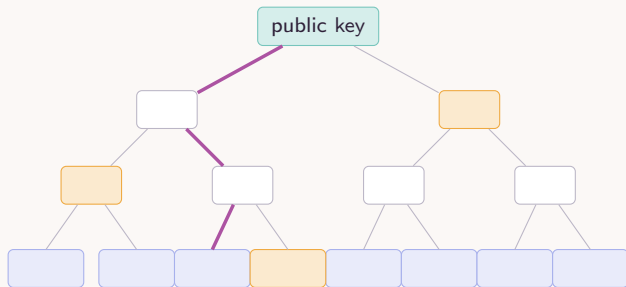
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Public key and signature together fit in a tweet.
- Signing is slow, constant-time implementation is still open, and the security assumption is the youngest of the lot.
- CSIDH is vulnerable to Kuperberg
- SIDH, the previous favourite, was broken in an afternoon in 2022.

You young people with your fancy math!
We know how to build cryptography.

Symmetric: One way to signatures



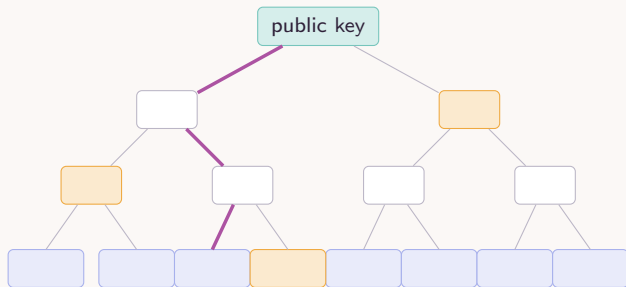
a few thousand one-time keys

siblings: the rest of the path

the signature is this path

one one-time key, used once

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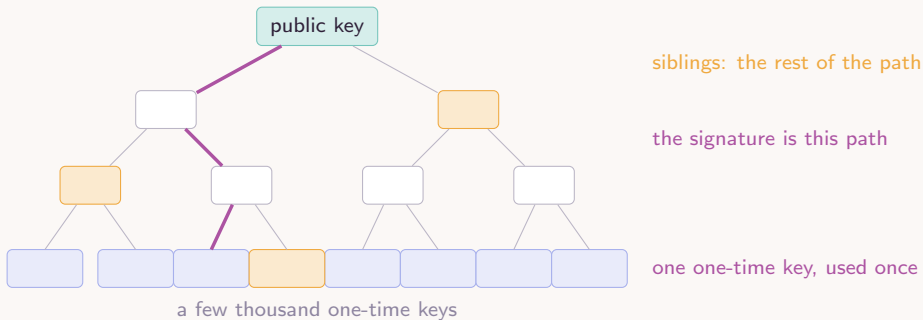
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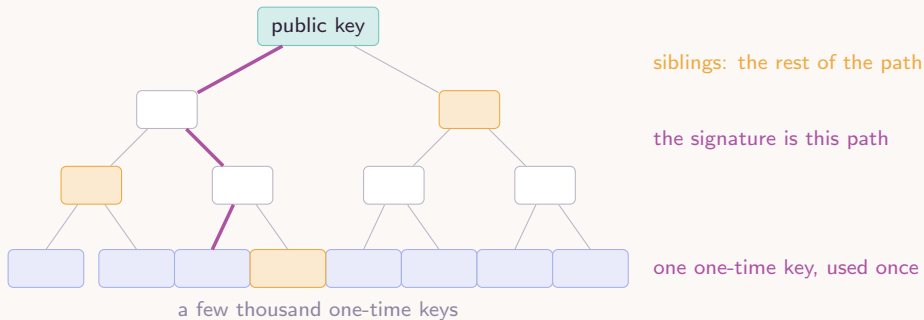
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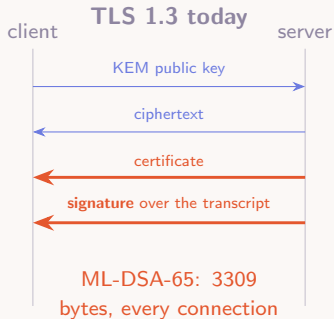
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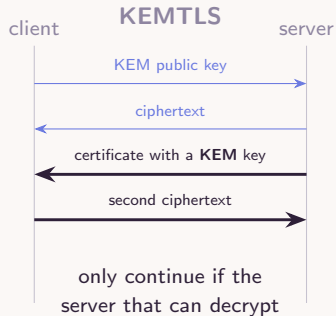
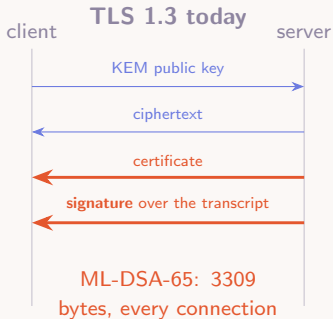
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- Security reduces to underlying hash function. We need one anyways, somewhere

But signatures are expensive!

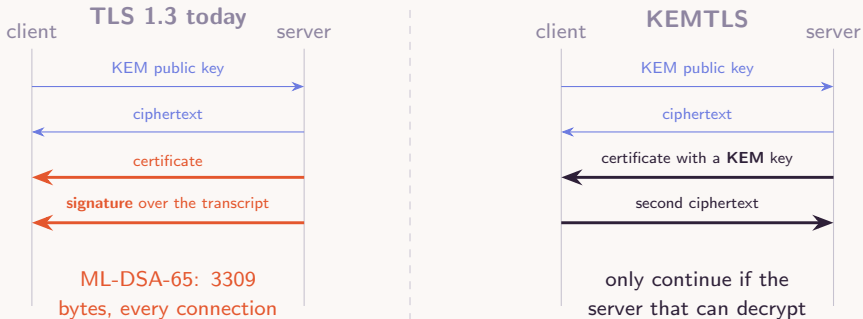
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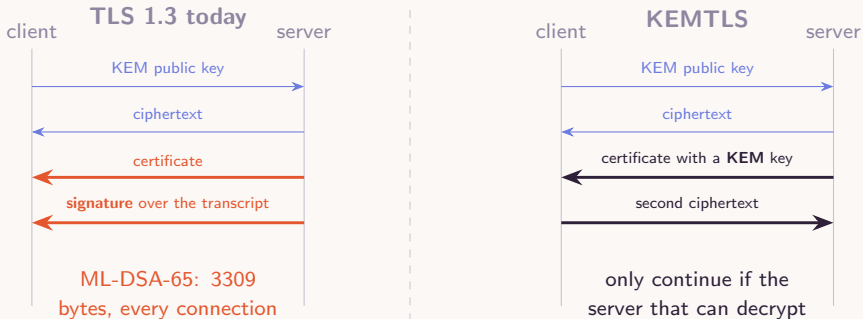


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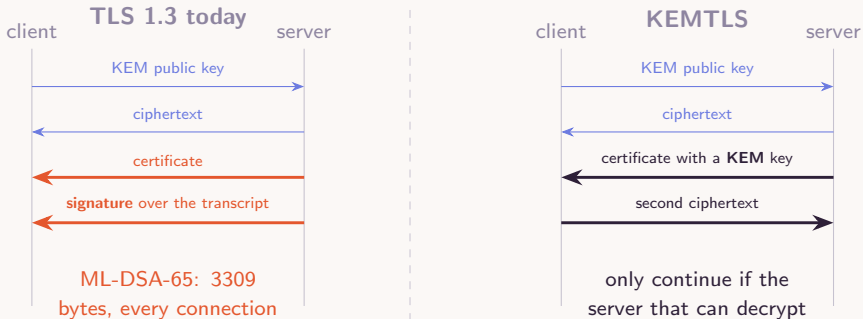
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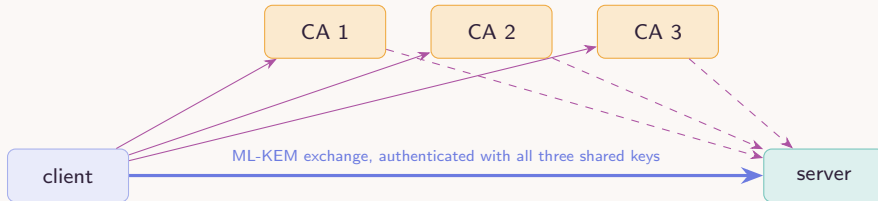
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- The CA still signs your certificate. KEMTLS removes the one signature, MTCs remove the four made in advance.

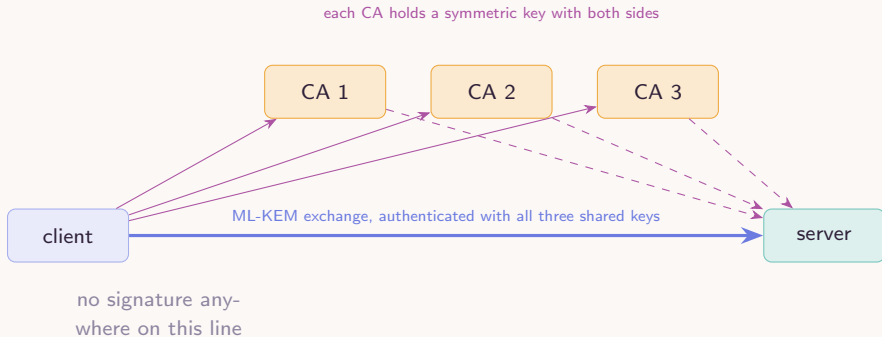
Down the symmetric rabbithole

each CA holds a symmetric key with both sides



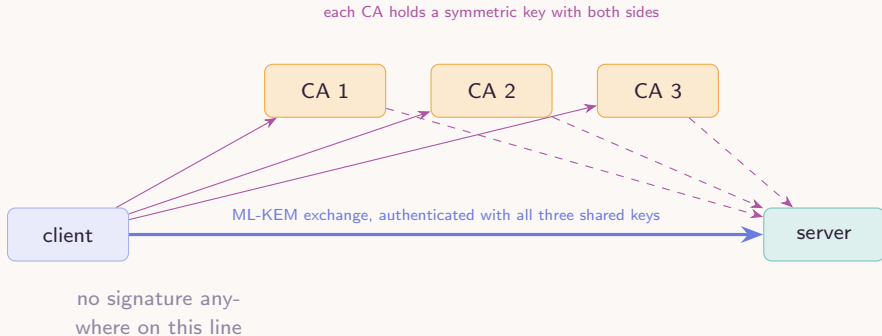
no signature anywhere on this line

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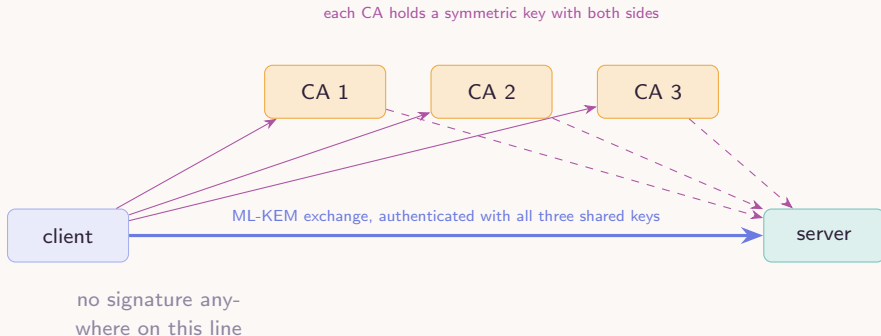
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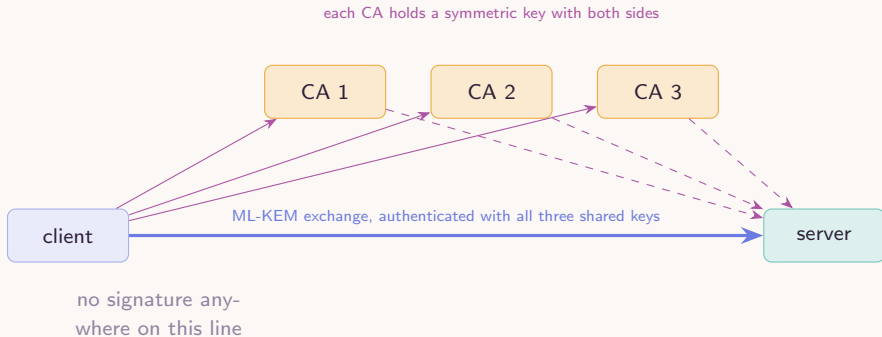
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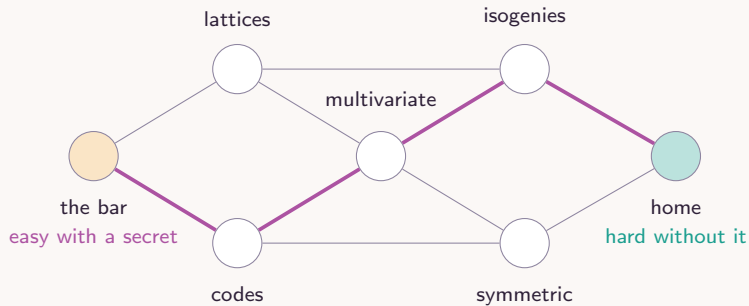
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Adam Langley, <https://www.imperialviolet.org/2024/04/07/letskerberos.html>

Her walk, in hindsight



Where do we go?



the Deployer

“The ethical stage: committing to an algorithm”

Start where you are

standard	what it is	status
ML-KEM (FIPS 203)	key exchange, lattices	shipping, everywhere
ML-DSA (FIPS 204)	signatures, lattices	the default
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- That's it.
- Everything else is either research prototype or in standardization process.
- You go to war with the algorithms you have, not the ones you wish you had.

<https://educatedguesswork.org/posts/pq-emergency/>, Eric Rescola

- The internet has decided. Browsers, OpenSSL, Go, Apple, the big CDNs: post-quantum key exchange is on by default and the share only goes up.

Nobody asked you

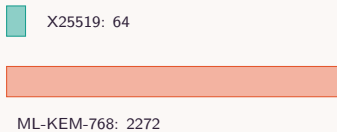
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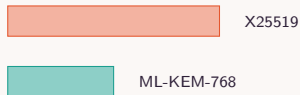
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- Betting against engineering problems usually means losing.

But it is bigger

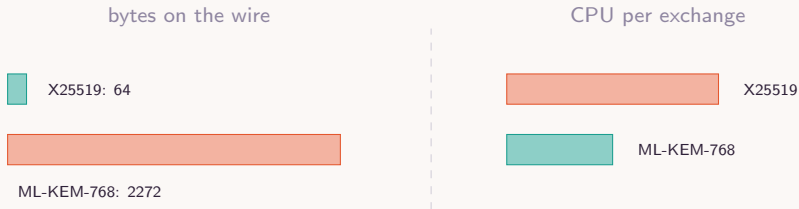
bytes on the wire



CPU per exchange

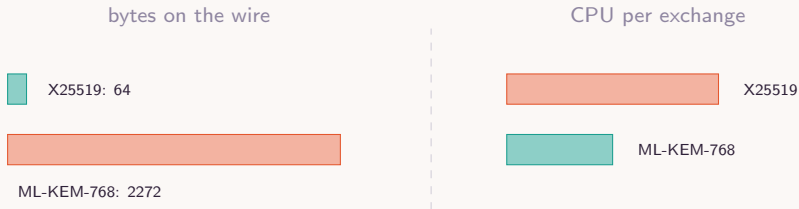


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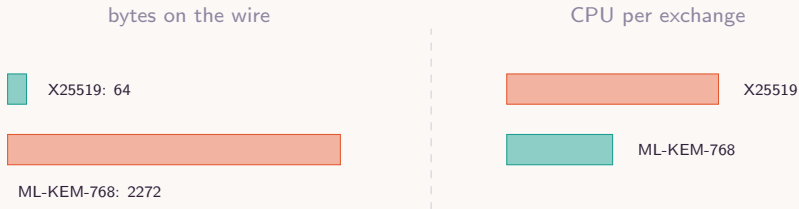
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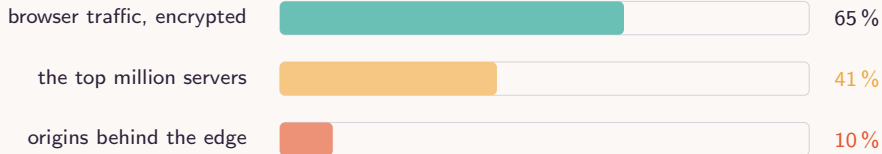
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- Lattices are bigger, but **faster** than elliptic curves.
- bytes are in the hot path of every connection, including a bad mobile link
- if you ship an Electron app tho...

The state of the post-quantum internet

done

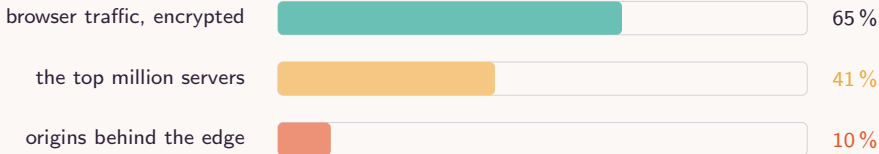


not started

FN-DSA still draft · hybrid certificate format still unsettled · Merkle tree certificates in testing

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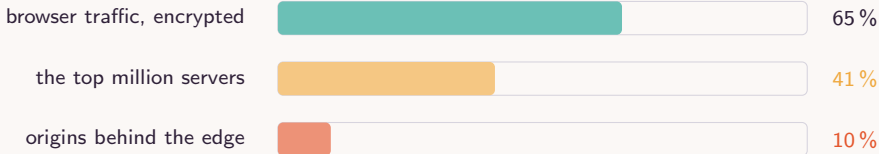
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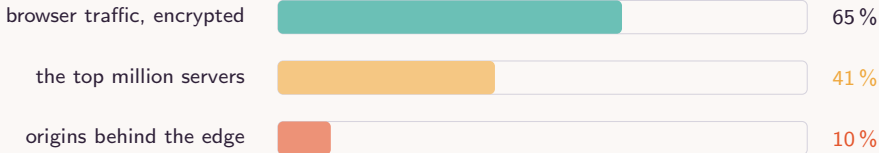
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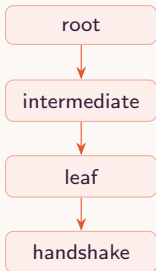
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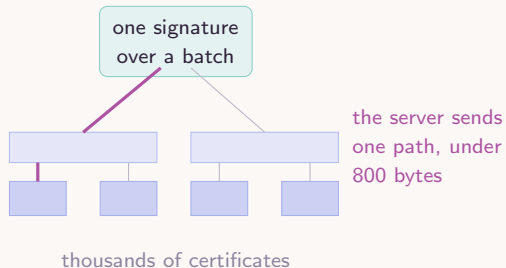
- Encryption is nearly done, took ten years, most middleboxes.
- Authentication is experimental.
- Cloudflare and Tranco scans, mid-2026.

If it can be broken, it can be fixed.

Merkle Tree Certificates

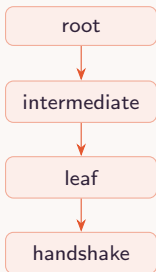


five signatures, sent every time

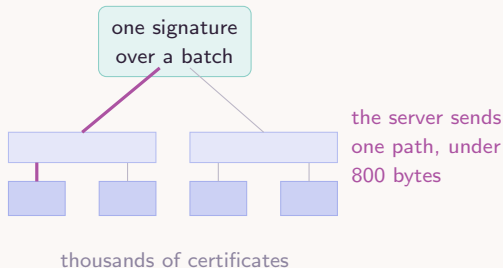


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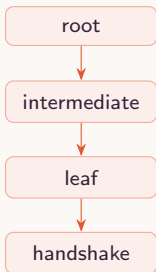


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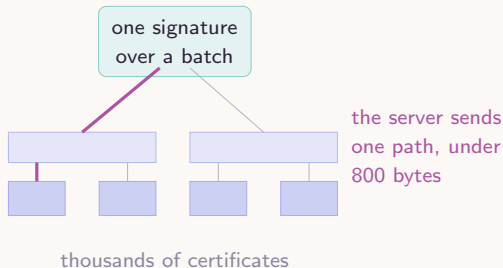


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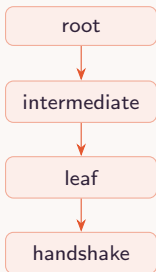


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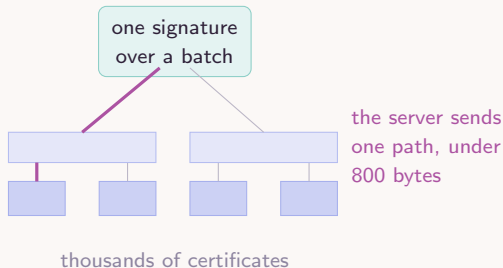


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- trust moves from “verify a chain now” to “I already know the root”.
- PQ+auth could be faster than classical

When does a useful quantum computer arrive?

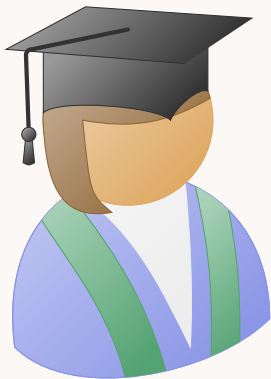
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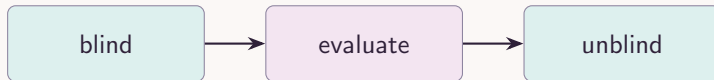
- Estimates from the early 2030s to the mid-2040s
- A lot of arguments: Hybrids, key sizes, which curve to keep
- Transition is as soon as 2029 (Google, Cloudflare)



the Academic

“What is to be done?”

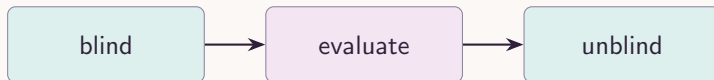
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$H(x)^r$, then $(\cdot)^k$, then take the r -th root

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- Exact, leak-free blinding: Kuperberg.
- Need to slap on a lot of zero-knowledge proofs and counting arguments.

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- Distinguisher Classic McEliece, became a quasi-polynomial key recovery.

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Math is beautiful, we get to open systems nobody has opened in thirty years, and not everything is decided.

Stop being sad.

Break something. On a laptop. Over a weekend.

Stop being sad.

Break something. On a laptop. Over a weekend.

Or fix something.

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Thank you.

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